# Zehan Ruan

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## SUMMARY OF QUALIFICATIONS

- Creativity developed in Building an AI visual obstacle avoidance car.
- Electronics & programming skills via self-built drone, AI car, rocket (PBL projects).
- Theoretical knowledge: AMC12 Distinction, physics/chemistry awards, C++ OI.
- Electrical circuit experience shown on Self-made FPV drone.
- Resource searching ability shown on finishing two PBL without any teachers' help and got 5 on 7 AP exam with self-study.
- Problem-solving: Overcame hardware failures in drone/rocket builds.
- Language: English, Chinese, Spanish

## EXPERIENCE

**Teacher**, Students' parents, Wenzhou City, Zhejiang Province, China March-April 2026

- Accessed to the student's education system to provide a learning method that matched with both of this education rules and student's learning habit, so that my class could reach to the highest efficiency.

**Dessert shop's manager**, Ruian High School, Wenzhou City, Zhejiang Province, China October 2025

- Did a survey about which kind of dessert is the most popular one from the students
- According to the survey's result, prepared the dessert's material and made them up to a half-processed dessert in advance
- Divided jobs and assigned them to my workers

**Tech Club Founder &President,** Ruian High School, Wenzhou City, Zhejiang Province, China (2023-2026)

• Led Raspberry Pi, drone, rocket, Android app projects; inspired student game dev.

**PBL Project Lead – Custom Drone,** Ruian High School, Wenzhou City, Zhejiang Province, China (Grade 11-12)

• Designed, soldered, coded, tuned quadcopter (PID, GPS, FPV); overcame crashes/explosions.

**PBL Project Lead – AI Avoidance Car & Rocket,** Ruian High School, Wenzhou City, Zhejiang Province, China (2024-2025)

• Built embedded car with AI recognition; synthesized rocket fuel, Arduino data logging.

**Android App Developer (Tomato Clock),** Personal (2024)

• Developed cross-device to-do app with server simulation.

## EDUCATION

**Candidate for Bachelor of Applied Science in Electrical Engineering**
September 2026-present
University of Waterloo, Waterloo, Ontario

**High School Diploma** June 2026
Ruian High School, Wenzhou City, Zhejiang Province, China

**Relevant Courses**: Physics, Calculus, Chemical, Biology, Statistic, Computer Science

## RELEVANT PROJECTS

**Custom FPV Drone, PBL** (2024-2025)
• Independently assembled high-speed long-range quadcopter; PID tuning, wireless, night flight add-ons.

**AI Avoidance Smart Car, Qingdao AI PBL** (2024)
• Group leader: Python control, camera AI training (100 images), obstacle avoidance.

**KNSB Rocket Test Rig, PBL** (2025)
• Led fuel synthesis, 3D printed body, Arduino 80Hz pressure data logging.

**Android Tomato Clock App** (2024)
• Full stack: Data sync server (Linux sim), to-do list, notifications.

**Personal School Website** (2024)
• MySQL backend, AI summary charts for student scores.

**C++ Multiplayer SanGuoSha** (2023)
• Complex logic, networking.

## AWARDS AND CERTIFICATES

**American Mathematics Competitions 12 (AMC12), Mathematical Association of America**, USA 2024-2025
• Certificate of Distinction, 2nd Place, Excellence &AIME Qualifier; advanced math problem solving
**Euclid Mathematics Contest**, University of Waterloo, Canada Grade 11
• Global Top 25%; mathematical reasoning
**JCCO Canadian Chemistry Olympiad**, Canada Grade 10
• National Gold, Global Bronze; chemistry knowledge
**CCC Canadian Chemistry Contest**, Canada Grade 11
• Regional Excellence; advanced chemistry
**Australian Mathematics Competition (Kangaroo)**, Australia Grade 10
• Gold Award; logical thinking
**Australian Mathematics Competition (AMC)**, Australia Grade 10
• National First Prize; math proficiency
**British Physics Olympiad**, UK Grade 10
• Global Bronze; physics knowledge
**ACSL American Computer Science League**, USA Grade 11
• National Qualifier; CS &logic
**Zhejiang C++ Informatics Olympiad**, China High School
• Second Prize; programming skills
**AP Computer Science A**, College Board, USA High School

• Score 5; Java programming

## ACTIVITIES AND INTERESTS

- ***Harvard Summit (Computer Biology &Bioengineering)****, Harvard University, USA*
  *Summer 2023*
  *• Participant; explored brain-machine interfaces and bio-electronics*
- ***Kunshan Duke University Math Modeling Contest****, Kunshan Duke University, China*
  *2024*
  *• Competitor; solved Rubik's cube optimal steps using breadth-first search*
- ***Technology Club****, Ruian High School, Wenzhou, Zhejiang, China* *2023-2026*
  *• Founder &President; led drone, rocket, AI car, app development projects*